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Adaptive Multi-Agent Coordination among Different Team Attribute Tasks via Contextual Meta-Reinforcement Learning

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Abstract: In the realm of Multi-Agent Reinforcement Learning (MARL), the challenge of ensuring effective coordination in different multi-agent teams remains a significant hurdle. Existing methods often fall short in generalizing learned policies to novel team compositions, sizes, and capabilities. Addressing this gap, our study focuses on systems with variable and obscure attribute compositions, harnessing a context-based meta-reinforcement learning framework. The approach is twofold: context inference and context-based decision-making. Agents interpret historical data to identify the system's attribute composition, guiding their collective efforts. The accuracy of these inferences is crucial, prompting us to integrate contrastive learning to refine the context inference network via unsupervised training. In the process of decision-making, agents integrate the inferred context with the observation features to select the optimal strategy. Our empirical results underscore the method's efficacy in bolstering decision-making efficiency and precision amidst attribute diversity, marking a significant stride in adaptive teaming for robust multi-robot deployments in dynamic real-world scenarios.

Key Words: Multi-Agent Reinforcement Learning, Meta-Learning, Variable Team Attribute Tasks

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1 Introduction

In the dynamic nexus of artificial intelligence and advanced robotics, variable multi-agent teams are heralded as the cornerstone for addressing complex tasks across diverse domains, including environmental surveillance, urban logistics, and precision agriculture [1–3]. The effective deployment of these teams pivots on the intricate dynamics of communication, coordination, and adaptive control, necessitating sophisticated mechanisms that can seamlessly navigate the multifaceted challenges presented by varied operational environments [4, 5].

In recent times, Reinforcement Learning (RL) has emerged as a powerful paradigm, driving significant advancements in the domain of autonomous agents and robotics [6–8]. By leveraging the potential of RL, agents can autonomously learn optimal behaviors through interaction with their environment, paving the way for solutions that are both robust and scalable [9, 10]. In this framework, the focus of MARL extends to controling multi-agent systems to collaborate in accomplishing tasks. MARL has led to remarkable achievements in fixed multi-agent systems, where each agent possesses fixed capabilities and attributes [11].

However, the application of these MARL methodologies often hits a bottleneck when introduced to variable multi-agent systems. These systems, characterized by diversity in capabilities, roles, and objectives, demand a level of flexibility and adaptability that goes beyond these MARL frameworks. For instance, if the detection range of an unmanned aerial vehicle is altered due to interference, it is imperative to ensure maximal area coverage by the swarm. This necessitates that the multi-agent system adaptively makes decisions based on the composition of its attributes. Current approaches tend to treat each agent as an isolated case, necessitating the design of individual models tailored for each unique agent configuration [12, 13]. This approach not only lacks scalability but also falls short in providing the dynamism required for agents to proficiently handle unforeseen changes in team composition or task requirements, a common occurrence in real-world applications.

To surmount these challenges, we introduce the concept of Meta-Learning, or 'learning to learn', as a transformative approach to equip multi-agent teams with the ability to swiftly adapt and evolve [14]. At its core, Meta-Learning transcends the confines of conventional learning paradigms by enabling systems to leverage prior learning experiences, thereby expediting the learning process for new and diverse tasks [15, 16]. This is particularly beneficial in the context of variable teams, where the ability to rapidly assimilate new information and adjust strategies in response to dynamic operational landscapes is paramount. By incorporating Meta-Learning, our framework aims to mitigate the rigidity and enhance the generalizability of current models. This approach contributes to the development of multi-agent systems that are more adaptive and resilient, better suited to manage the complexities and unpredictability of real-world scenarios.

Specifically, we consider the attribute composition in multi-agent systems as the context within the Meta-Learning framework. We then infer this context from historical trajectories of interaction with the environment, utilizing it for decision-making purposes. In this process, the quality of context inference is crucial. Therefore, we employ contrastive learning methods to aid in the training of the context inference module. The purpose of contrastive learning here is to enhance the discriminative ability of the model, enabling it to more accurately differentiate between varying contexts and improve its overall effectiveness in decision-making scenarios. Our contributions are summarized as follows:

1) By leveraging Meta-Learning, we propose a MARL

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framework suited for variable multi-agent systems.
2) Utilizing contrastive learning, we enhance the context inference performance within Meta-Reinforcement Learning.
3) Experiments conducted in both simulated and real-world settings validate the efficacy of our approach and its superiority over baseline methods.

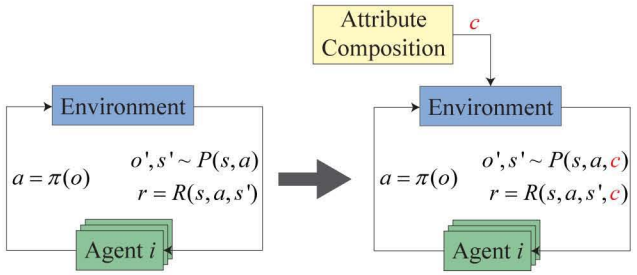
2 Related Work

2.1 Multi-Agent Reinforcement Learning

In recent years, the field of MARL has advanced rapidly, with Centralized Training with Decentralized Execution (CTDE) emerging as a particularly prominent framework [17–20]. CTDE adeptly balances individual autonomy and collaborative efficacy, allowing each agent to make independent decisions based on local observations while ensuring cohesive global cooperation. Among the methodologies within CTDE, value-based approaches like Value Decomposition Networks (VDN) [21] and QMIX [22], along with their subsequent enhancements, have been recognized for effectively decomposing the global value function into agent-specific components. This decomposition simplifies the complex task of assigning credit among agents, a crucial aspect of learning in multi-agent environments. Parallel to this, policy-based methods such as Multi-Agent Proximal Policy Optimization (MAPPO) [20] and Multi-Agent Soft Actor-Critic (MASAC) [23] have gained attention. These approaches integrate the global state into the Critic network during training, facilitating centralized training. This centralization is key to fostering a higher degree of cooperation among agents, enhancing the overall performance of the multi-agent system.

2.2 Generalization of MARL

The generalization of MARL presents a promising area of application. Prior works in MARL have predominantly focused on fixed multi-agent systems. Some role-based MARL research has concentrated on the diverse behaviors exhibited by agents [24, 25]. A number of studies have achieved decision-making in variable multi-agent systems by sharing parameters and appending agent IDs to observations [11, 26]. However, this approach suffers from limited generalization, only functioning effectively within a predefined set of multi-agent configurations. CACAT [27] introduces a method of incorporating specific capability values into observations to facilitate decision-making among variable multi-agent teams, a technique that might be strong in theory but is potentially fraught with assumptions in practical scenarios. This is because, in dynamically evolving multi-agent systems, the attributes (or capabilities) of agents can be volatile and may not always provide accurate values in real-time. We introduce meta-reinforcement learning as a robust solution to this challenge. Meta-reinforcement learning adapts effectively to the dynamic nature of multi-agent systems, offering a flexible and potent approach to handle the instability and uncertainty of agent capabilities, thereby addressing the limitations of previous methods in capturing the complexities of real-world scenarios.



(a) Typical Dec-POMDP (b) Contextual Dec-POMDP

Fig. 1: Comparison of typical Dec-POMDP and Contextual Dec-POMDP.

3 Preliminaries

3.1 Contextual Decentralized Partially Observable Markov Decision Processes

Our multi-agent problems can be formulated as Contextual Decentralized Partially Observable Markov Decision Processes (Dec-POMDPs) [28, 29]: $\langle \mathcal{C}, S, A, O, P, R, \gamma \rangle$. $\mathcal{C}$ is the set of contexts, where each context $c \in \mathcal{C}$ corresponds to a different multi-agent attribute composition in our problem. S represents the set of global states in the environment. Each state $s \in S$ encapsulates the comprehensive status of all agents and relevant environmental factors at a given time. In Dec-POMDPs, the global state is not fully observable to each agent, which introduces partial observability into the decision-making process. A denotes the set of joint actions available to all agents. An action $a \in A$ involves the collective decisions made by all agents at a particular time step. This set encompasses all possible combinations of individual actions a_i that can be taken by the agents in the system. O is the set of observations for the agents. Each observation $o_i \in O$ represents a partial view or perception of the global state s, as perceived by an individual agent i. This reflects the decentralized nature of information in Dec-POMDPs, where each agent only has a limited and local perspective of the entire state space. P is the state transition function specifying the probability $P(s'|s, a, c)$. R is the reward function which maps the global state and joint actions to a single scalar reward $r = R(s, a, c)$. Unlike typical Dec-POMDPs, transition and reward functions are dependent on context c. Within each multi-agent attribute composition, the agents are expected to make cooperative decisions to maximize the sum of future rewards.

3.2 Multi-Agent Soft Actor-Critic Algorithm

MASAC algorithm extends the Soft Actor-Critic (SAC) [30] framework, a popular off-policy actor-critic deep reinforcement learning algorithm, to multi-agent environments. MASAC is designed to efficiently handle the complexities involved in learning policies for each agent, considering the presence and potential strategies of other agents in the environment. The core of MASAC involves each agent learning a policy that maximizes not only the expected cumulative reward but also the entropy of the policy. This encourages exploration and prevents premature convergence to suboptimal policies. The learning process involves two main components for each agent: an actor network for policy approximation and a critic network for value function approxima-

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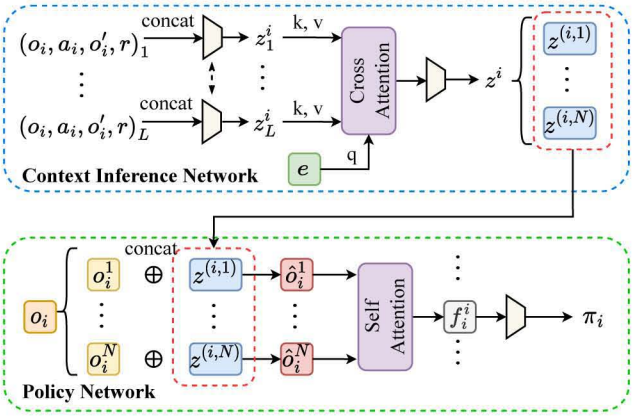


Fig. 2: The network architecture of the context inference module and the policy module.

tion. The critic network is updated to minimize the Bellman error, and the actor network is updated to maximize a trade-off between expected reward and policy entropy. The critic is updated by minimizing the loss function:

$$\mathcal{L} = \mathbb{E}_{(s,a,r,s')\sim\mathcal{D}}\left[\sum_i \left(Q_i(s,a_i) - y_i(r,s')\right)^2\right] \quad (1)$$

$$y_i(r,s') = r + \gamma\left[Q_i'(s',a_i') - \alpha \log \pi_i(a_i'|o_i')\right] \quad (2)$$

where α is the temperature parameter that determines the relative importance of the entropy term versus the reward. The actor network parameters are updated by performing gradient ascent on the objective:

$$\mathcal{J} = \mathbb{E}_{s\sim\mathcal{D},a_i\sim\pi_i}\left[\sum_i \left(Q_i(s,a_i) - \alpha \log \pi_i(a_i|o_i)\right)\right] \quad (3)$$

In both updates, $\mathcal{D}$ denotes the replay buffer that stores the experience tuples, Q_i represents the critic network for agent i, and π_i represents the policy network (actor) for agent i. The delayed policy updates in MASAC help in stabilizing the training by reducing overestimation bias and smoothing the policy update.

4 Generalized MARL via Meta-Learning

We achieve generalization in MARL by introducing a meta-reinforcement learning approach. Our algorithm is implemented based on MASAC, hence we refer to our method as Attribute Generalized MASAC, abbreviated as AG-MASAC.

4.1 Context Inference and Context-Based Decision-Making

We address the decision-making problem under varying attribute compositions through context-based meta-reinforcement learning. We decompose this problem into two parts: context inference and decision-making based on context. These correspond to the context inference module and the context-based decision network, respectively. The context inference module deduces the attribute composition of the multi-agent system from the historical trajectories, and the network structure is illustrated in Fig. 2.

Suppose the number of agents is N. Initially, agent i employs an encoder network composed of a Multi-Layer Perceptron (MLP) to infer the latent variables of the entire multi-agent system's attribute composition, $z_l^{(i,\{1,\ldots,N\})} = (z_l^{(i,1)},\ldots,z_l^{(i,N)})$, from transition samples $(o_i,a_i,o_i',r)_l$. Subsequently, all samples are aggregated via a cross-attention module and a MLP to obtain more precise inferred features $z^{(i,\{1,\ldots,N\})}$. Within the attention mechanism, a learnable variable e serves as the query vector, while the latent variables of the multi-agent attribute composition, inferred from the samples, act as the key and value vectors. The computation of the attention mechanism is formulated as shown in Equation (4).

$$z^{(i,\{1,\ldots,N\})} = \text{softmax}\left(\frac{e(z_{\{1,\ldots,L\}}^{(i,\{1,\ldots,N\})})^T}{\sqrt{d_z}}\right) z_{\{1,\ldots,L\}}^{(i,\{1,\ldots,N\})} \quad (4)$$

where d_z represents the dimensionality of the key vectors.

After inferring the attribute composition of the entire multi-agent system, agents utilize this information for decision-making. In the context-based decision network (illustrated in Fig. 2), the observation of agent i, o_i, is decomposed into observation features o_{ij} for different enenties in the environment. The inferred latent attribute variable $z^{(i,j)}$ of each agent is then concatenated with the corresponding observation feature. For entities that do not require inference (like resource point), a zero vector is concatenated instead. Subsequently, all observation features are processed through a self-attention block to yield the decision variable.

$$f_i = \text{softmax}\left(\frac{\hat{o}_i\hat{o}_i^T}{\sqrt{d_{\hat{o}_i}}}\right)\hat{o}_i \quad (5)$$

It's important to note that the number of output vectors from the self-attention block is equal to the number of input vectors. For the ith agent, the observation feature o_i^i represents its own features. After processing through the self-attention mechanism, this vector incorporates information from all other entities, hence the ith vector f_i^i is selected as the decision vector. Then, the policy network computes the action distribution $\pi_i(a_i|o_i, z^{(i,\{1,\ldots,N\})}) = \text{MLP}_\pi(f_i^i)$. The action-value network calculates $Q_i(s,a_i,z^{(i,\{1,\ldots,N\})}) = \text{MLP}_Q(s,f_i^i,a_i)$. Herein, the global state s is utilized solely during training to compute more accurate action-values, whereas only local observations are required during the multi-agent decision-making phase.

4.2 Enhanced Context Representation Based on Contrastive Learning

In the context inference module, prior experience data is utilized to infer the attribute composition of the multi-agent system, which is then used for decision-making. As such, the inferred attribute representation significantly influences performance, necessitating the enhancement of the quality of context inference. We employ contrastive learning in the context inference module of meta-reinforcement learning, focusing on the contrasting nature behind different attributes and utilizing the unsupervised training of the context inference network through contrastive learning. By labeling experience data from the same attributes as positive samples

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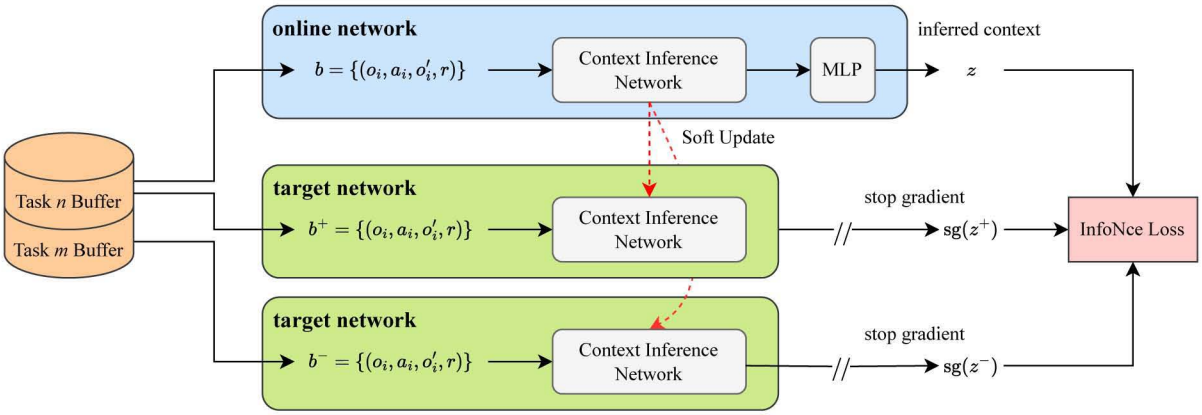


Fig. 3: Framework of contrastive learning.

and computing the contrastive loss through the asymmetric structure of online and target networks, we minimize the contrastive loss to generates high-quality latent variables for the multi-agent system's attribute composition, ensuring that the features of different attribute compositions are more distinct while those of the same attribute composition are more similar. Two neural networks are used for learning: an online network and a target network. The parameters of the online network are denoted by ϕ, consisting of a context inference network and a final MLP layer, as shown in Fig. 3. The target network has the same context inference network structure as the online network but uses different parameters ξ and does not include the final MLP layer.

Specifically, for each attribute composition, transition sample data is obtained through interaction and stored in the corresponding experience replay pool. During training, for each experience replay pool, two sample sets, b and b', are sampled separately. z and z^+ are obtained as positive samples through the online and target networks, respectively. We obtain some transition samples from the replay pool of other attributes, and negative samples z^- are acquired through the target network. The InfoNCE loss is a contrastive loss function commonly used in self-supervised learning, which is designed to bring similar samples closer and push dissimilar samples further apart in the feature space:

$$\mathcal{L}_{\text{con}} = -\log \frac{\exp(\text{sim}(z, \text{sg}(z^+))/\tau)}{\sum_{z^-} \exp(\text{sim}(z, \text{sg}(z^-))/\tau)} \tag{6}$$

where sim() represents the cosine similarity function, sg() means the stopping of gradient and τ is the temperature coefficient. The target network is iteratively updated through a soft update process; every fixed number of training steps, the parameters of the online network are weighted and copied to update the parameters of the target network:

$$\xi \leftarrow \beta\xi + (1-\beta)\phi \tag{7}$$

where $0 < \beta < 1$ is a hyperparameter.

4.3 Overall Training Procedure

Meta-Reinforcement Learning is divided into two phases: meta-training and meta-testing, detailed respectively in Algorithms 1 and 2. The process alternates between meta-training and meta-testing, conducting a meta-test at specific intervals during the meta-training phase (every certain number of steps) and using the performance of the meta-test as an evaluation metric for the model.

During the training phase, the policy network parameters, θ, and the action-value network parameters, ψ, are updated using the loss modified from the MASAC algorithm:

$$\mathcal{L}_{\text{act}}(d, z) = \sum_{(s,a,r,s')} \sum_{i} \left(\alpha \log \pi_{\theta_i}(a_i|o_i, z^i) - Q_{\psi_i}(s, a_i, z^i) \right) \tag{8}$$

$$\mathcal{L}_{\text{cri}}(d, z) = \sum_{(s,a,r,s')} \sum_{i} \left(Q_{\psi_i}(s, a_i, z^i) - y_i(r, s', z^i) \right)^2$$

$$y_i(r, s') = r + \gamma \left[Q_{\psi_i'}(s', a_i', z^i) - \alpha \log \pi_{\theta_i}(a_i'|o_i', z^i) \right] \tag{9}$$

where d represents the set of transition tuples, and z denotes the inferred context. The online network within the context inference module is trained through the loss derived from contrastive learning, followed by training the target network via a soft update mechanism. Additionally, the critic loss in reinforcement learning is also utilized to train the context inference module, to foster the learning of context representations that enhance the performance of the agents.

5 Simulated Experiments

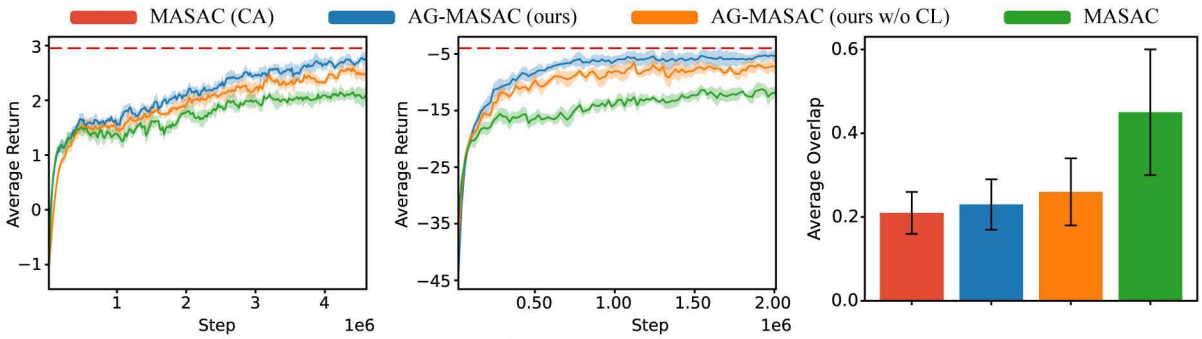
5.1 Experimental Setup

We conduct experiments in two multi-agent cooperative environments with attribute generalization: Heterogeneous Sensor Network (HSN) and Heterogeneous Collection (HC). In the HSN environment, each agent has a different monitoring radius (equal to the connection radius), and the agents aim to maximize the coverage area while maintaining connectivity. In the HC environment, each agent possesses a distinct maximum speed, and the agents strive to collect as many resource points as possible within a limited timeframe. The attribute composition within each episode environment varies, for instance, a radius combination of (1, 2, 3) may change to (0.4, 2, 2.3), prompting the agents to adapt their cooperative behavior based on the attribute composition. Visualizations of the two environments are shown in Fig. 5.

5.2 Experiment Result and Analysis

In meta-reinforcement learning, the performance during meta-testing is utilized as an evaluation metric. During meta-testing, a batch of attribute compositions is sampled to serve

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(a) HC: Average Return (b) HSN: Average Return (c) HSN: Average Overlap

Fig. 4: Experimental Results: Performance on the training set. (a) represents the return curve on HC (higher is better), (b) denotes the return curve on HSN (higher is better), and (c) shows the overlap area on HSN after completing training (lower is better).

Algorithm 1 AG-MASAC: Meta-Training

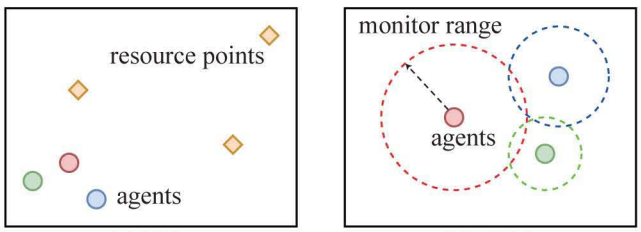
Input: A set of multi-agent attributes for training, denoted as $\mathcal{C}_{\text{train}} = \{c\}$

Initialize an experience replay pool $\mathcal{D}_c$ for each c.
while not converged **do**
  for $c \in \mathcal{C}_{\text{train}}$ **do**
    Initialize a sample set $b_c = \{\}$ for context inference.
    for number of rollout episodes **do**
      Agent i infers context $z^i = q_\phi(b_c)$.
      Collect transitions using the policy $\pi_{\theta_i}(a_i|o_i, z^i)$ and store it in $\mathcal{D}_c$.
      Sample some transitions from $\mathcal{D}_c$ to update b_c.
    end for
  end for
  for $c \in \mathcal{C}_{\text{train}}$ **do**
    Sample some transitions b_c and d_c, from $\mathcal{D}_c$.
    Agents infer the context $z = q_\phi(b_c)$.
    Calculate RL training losses $\mathcal{L}^c_{\text{act}} = \mathcal{L}_{\text{act}}(d_c, z)$, $\mathcal{L}^c_{\text{cri}} = \mathcal{L}_{\text{cri}}(d_c, z)$ using Equation (8) and (9).
  end for
  Calculate contrastive losses $\mathcal{L}_{\text{con}}$ using Equation (6).
  $\phi \leftarrow \phi - \alpha_1 \nabla_\phi (\sum_c \mathcal{L}^c_{\text{cri}} + \alpha_2 \mathcal{L}_{\text{con}})$
  $\theta \leftarrow \theta - \alpha_3 \nabla_\theta \sum_c \mathcal{L}^c_{\text{act}}$
  $\psi \leftarrow \psi - \alpha_4 \nabla_\psi \sum_c \mathcal{L}^c_{\text{cri}}$
end while

Algorithm 2 AG-MASAC: Meta-Testing

Input: A set of multi-agent attributes for testing, denoted as $\mathcal{C}_{\text{train}} = \{c\}$

for multi-agent attribute composition $c \in \mathcal{C}_{\text{train}}$ **do**
  Initialize a sample set $b_c = \{\}$ for context inference.
  for number of testing episodes **do**
    Agent i infers context $z^i = q_\phi(b_c)$.
    Collect transitions using the policy $\pi_{\theta_i}(a_i|o_i, z^i)$ and store it in b_c.
  end for
end for



(a) HC (b) HSN

Fig. 5: Visualization of the two experimental environments (a) Heterogeneous Collection and (b) Heterogeneous Sensor Network.

as the testing set. If the training set itself is used for testing during meta-testing, it represents the performance on the training set. If a new batch of attribute compositions is resampled during meta-testing to form a test set, then it represents the performance on new attribute compositions. The performance of the proposed method on the training set is illustrated in Fig. 4. From the results, it can be observed that our method demonstrates superior performance in both experimental environments. MASAC can only learn an average policy under different attribute compositions, converging to a sub-optimal strategy. Our method, through inferring attributes from historical trajectories and making decisions based on inferred latent variables, can adjust the policy among different attribute compositions, resulting in better performance. It is evident that the performance of our method is close to MASAC (Composition-Aware, CA) which is achieved with knowledge of system attributes. This further suggests that we have successfully inferred effective information from historical trajectories. Additionally, Ours (w/o CL) represents the performance of our method after removing the contrastive learning loss. It can be observed that there is a certain loss in performance, yet it still significantly exceeds that of the baseline methods. This indicates that contrastive learning is beneficial in assisting agents to distinguish different attribute compositions.

5.3 Generalization on New Attribute Composition

To verify the generalizability of our method across different multi-agent attribute compositions, we resample the test set with new attribute distributions and add experimental scenarios with 4 and 5 agents, the results of which are illustrated in the Fig. 6 and Fig. 7. It is apparent that our method still performs well with new attribute compositions, significantly outperforming the baseline methods. This is because the context-based meta-reinforcement learning enables the agents to possess a universal strategy across different environments. In new attribute compositions, they can

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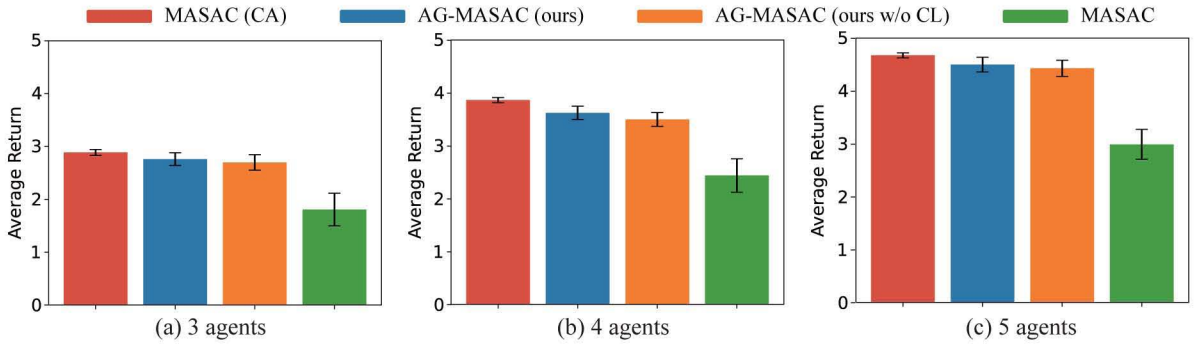


Fig. 6: Experimental results on the HC test set, where (a) to (c) represent the setups with 3-5 agents, respectively.

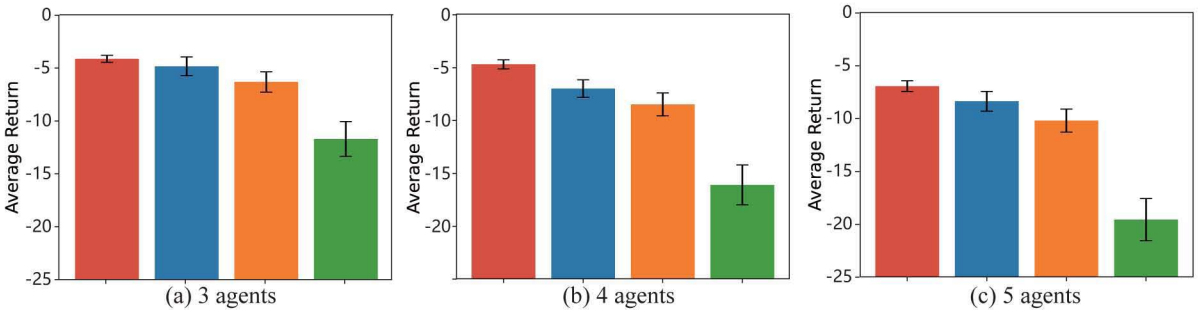


Fig. 7: Experimental results on the HSN test set, where (a) to (c) represent the setups with 3-5 agents, respectively.

immediately infer the new context through interaction with the environment and make appropriate decisions. In contrast, the baseline methods fail to perceive such changes and employ the same strategy across all scenarios, which proves to be inadequate in adapting to new environmental variations. In scenarios with an increased number of agents, our method remains applicable. This is because, after capturing the attribute information of each agent, the decision module's attention mechanism can calculate the interrelations with other agents, thereby facilitating decision-making. This process is independent of the number of agents.

5.4 Visualization of Inferred Context

To better understand the meaning of context inference, we sample some context inference variables and visualize them through dimensionality reduction, as shown in the Fig. 8. It can be observed that in both environments, our context inference latent variables have a relatively clear distribution relationship with the attributes of the agents. This indicates that we have indeed inferred their corresponding attributes from the historical trajectories of different attributes.

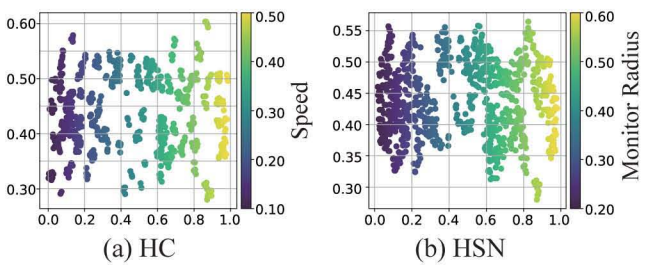


Fig. 8: Dimensionality reduction visualization of the context latent variables.

6 Real-World Experiments

Using models trained with reinforcement learning to control robots in accomplishing tasks in the real world is promising [31]. To validate the effectiveness and superiority of our proposed method in the real world, we conduct experiments based on a physical multi-robot system.

6.1 Experimental Setup

We employ the RoboMaster EP[1] as the physical robot, with size of approximately 320 mm in length, 240 mm in width, and 270 mm in height. This robot is equipped with Mecanum wheels, which facilitate multidirectional movement including forward, backward, left, and right. To gather essential data for robots' decision-making, we employ a motion capture system comprising cameras and reflective markers attached to the robot. These markers, small spheres with a reflective coating, enable the cameras to capture the robot's position and velocity efficiently. The robot and the motion capture system are shown in Fig. 9.

In the training process within the simulation environment, we save the policy network models from four different training steps of each environment and then deploy it on the robots to test its performance in real-world scenarios. We transmit observations to the robots at a frequency of 10 Hz. After receiving these observations, the robots compute and execute actions based on the deployed strategy to accomplish tasks.

6.2 Experimental Results and Analysis

For each model, we repeated experiments 10 times, with each experiment involving the random sampling of 5 distinct multi-agent attribute compositions. The results are shown in

[1] https://www.dji.com/global/robomaster-ep

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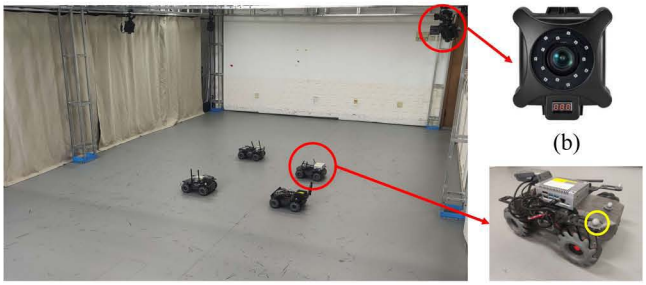


Fig. 9: The hardware platform for the multi-robot experiment. (a) is an overview of the entire experimental site, (b) is the camera of the motion capture system, (c) is the RoboMaster EP robot, where the yellow circles indicate reflective markers.

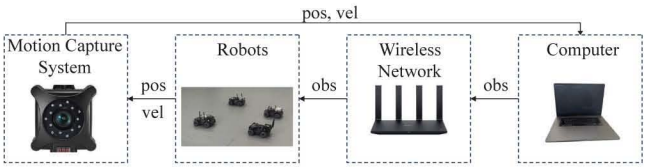


Fig. 10: The information flow framework of the multi-robot experiment.

Table 1 and Table 2. It can be observed that our method still performs excellently in the physical robot environment, significantly outperforming the baseline method MASAC. Comparing the performance after training completion, our method improved by 31.58% on HC and 43.51% on HSN over the baseline. This indicates that our method possesses strong adaptability and task completion performance when dealing with dynamic multi-agent problems in real-world scenarios. In contrast, the baseline method fails to adapt to the complex and ever-changing realities of real-world scenarios.

Table 1: The average return of the HC task in the physical experiment.

Train Steps	Ours	Ours (w/o CL)	MASAC
1M	**1.4**	1.4	1.1
2M	**1.7**	1.6	1.4
3M	**2.2**	2.0	1.7
4M	**2.5**	2.3	1.9

Table 2: The average return of the HSN task in the physical experiment.

Train Steps	Ours	Ours (w/o CL)	MASAC
0.5M	**-10.3**	-10.5	-16.2
1M	**-7.7**	-8.0	-14.7
1.5M	**-7.5**	-7.7	-13.5
2M	**-7.4**	-7.6	-13.1

6.3 Visualization Comparison

Fig. 11 provides snapshots at the end of some episodes in the physical experiment under two different multi-agent attribute compositions. It is apparent that in the HC environment, our method, due to its inference of the attribute composition of the multi-agent system, can make optimal task assignments, ultimately completing the collection of all resource points. In contrast, the baseline method lacks this capability, leading to a decrease in task completion efficiency. In the HSN environment, our method's agents achieve maximum coverage while ensuring connectivity with each other. Under the baseline method, agents fail to fully utilize a large detection radius, leading to wasted coverage area, and cannot maintain connectivity with a small detection radius.

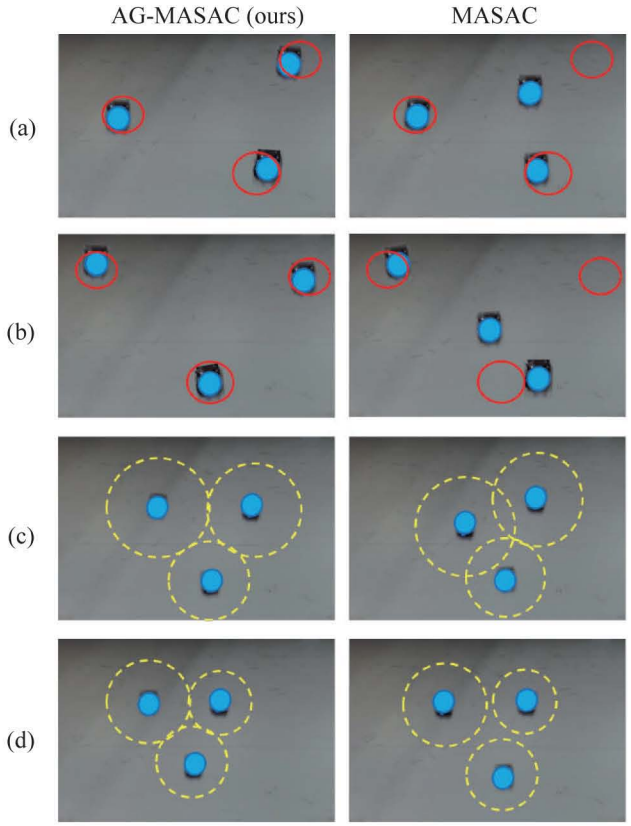


Fig. 11: Snapshots of the multi-robot experiment under different multi-agent attribute compositions. (a) and (b) show the HC environment with multi-agent velocity compositions of $(0.37, 0.62, 0.81)$ and $(0.34, 0.58, 0.83)$, respectively. Blue dots represent agents and red circles represent resource points. (c) and (d) show the HSN environment with multi-agent sensor radius compositions of $(0.7, 0.82, 0.9)$ and $(0.55, 0.62, 0.73)$, respectively. Blue dots indicate agents and yellow circles means monitor range.

7 Conclusion

In this paper, we implement high-generalization multi-agent reinforcement learning through meta-learning and supplement training with contextual representation via contrastive learning. Experimental results indicate that our approach can infer the attribute composition of multi-agent systems from historical trajectories, enabling targeted decision-making, with a performance exceeding baseline methods. However, the approach presented in this study may encounter limitations when dealing with extreme environmental changes or atypical data distributions. Future work will explore how to enhance the robustness of the model under these circumstances.

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